

MODULE 07

SQL Aggregate Functions

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Aggregate Functions

Aggregate Functions

COUNT(): count rows

SUM(): total of values

AVG(): average value

MIN(): minimum value

MAX(): maximum value

COUNT

```
SELECT COUNT(*) AS total FROM students;
SELECT COUNT(email) AS has_email FROM students;
SELECT COUNT(DISTINCT dept) AS depts FROM students;
```

SUM and AVG

```
SELECT SUM(salary) AS payroll FROM employees;
SELECT ROUND(AVG(gpa), 2) AS avg_gpa FROM students;
SELECT SUM(salary) FROM employees WHERE dept = 'Eng';
```

MIN and MAX

```
SELECT MAX(salary), MIN(salary) FROM employees;
SELECT MIN(enrollment_date) FROM students;
-- Name with highest GPA
SELECT * FROM students
WHERE gpa = (SELECT MAX(gpa) FROM students);
```

GROUP BY and HAVING

```
SELECT dept, COUNT(*), ROUND(AVG(gpa),2) AS avg
FROM students
GROUP BY dept
HAVING AVG(gpa) > 3.0
ORDER BY avg DESC;

-- WHERE filters rows (before grouping)
-- HAVING filters groups (after grouping)
```